MoveOn Phone Cover Downloader & Translation Pipeline

Version: 1.0 | **Project:** inventory/coverstore | **Status:** Approved

1. Goal & Architecture Overview

Automate the extraction of phone covers from **moveon.global** based strictly on user-supplied links. The tool identifies available phone models, auto-detects cover materials (with interactive prompt fallback), and downloads original showcase images and SKU variants into a structured local directory: \[phone model]\[material]\[original|sku|translated]. Non-English image text is translated to English using Gemini AI.

IMPORTANT SCOPE CONSTRAINT: Zero out-of-scope crawling. The tool strictly processes only the exact product URL provided by the user. No background scraping or related product fetching.

2. Local Folder Hierarchy

Directory Path	Contents
`\[phone model]\[material]\original\`	High-resolution product showcase and gallery images.
`\[phone model]\[material]\sku\`	Variant covers and color swatches matching selected phone model.
`\[phone model]\[material]\translated\`	Images with non-English text translated to English.
`\scripts\`	Python execution modules (downloader, client, material detector).
`\plan\plan.pdf`	This technical specification document.

3. MoveOn Integration Strategy

MoveOn's frontend is a Next.js application that calls an internal proxy API: https://product-api.proxy.service.moveon.global/api/product/customer/v1/products/find/{id}?region=BD&locale;=en&business;_type=wholesale. Direct API access with simulated guest session headers (X-GUEST-ID, X-GUEST-SESSION) retrieves full resolution images, attribute matrices (Color, Applicable Models), and SKU pricing without browser overhead.

4. Implementation Modules (in \scripts\)

Script	Responsibility
`scripts/moveon_client.py`	Parses MoveOn URLs, queries the product API, and maps models and SKUs.
`scripts/material_detector.py`	Classifies material from title/attributes (Silicone, TPU, Glass, Leather, etc.) with prompt fallback.
`scripts/downloader.py`	Interactive CLI runner: prompts for model, creates folders, downloads images with progress.
`scripts/translate_helper.py`	Detects non-English text on photos and coordinates translation into \translated\.

5. Verification & Workflow

1. Run python scripts/downloader.py --url <moveon_url>.
2. Tool inspects product title & attributes, auto-detecting the material (e.g. Liquid Silicone).
3. Tool prompts user with numbered list of available models.
4. Tool downloads gallery images into \[model]\[material]\original and matching variant covers into \[model]\[material]\sku.
5. Non-English images are sent to Gemini AI for translation into \[model]\[material]\translated.